

Recommendations — hgsoc-tuboovarian-stage3c-r0-hrd-pen ding-idyq

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — hgsoc-tuboovarian-stage3c-r0-hrd-pending

Profile snapshot

- **Primary site:** ovary
- **Histology:** high-grade serous carcinoma (tubo-ovarian / primary peritoneal; WT1+, PAX8+, SOX17+, CK7+)
- **Stage:** IIIc
- **ECOG:** 1
- **Age band:** 70-79
- **Sex:** female
- **Biomarkers:** TP53 V157G (somatic) somatic missense mutation in the DNA-binding domain; concordant with diffuse p53 IHC overexpression; BRCA1/2 (germline and somatic) wild-type — no pathogenic variant germline (74-gene panel) and no actionable somatic alteration on tumor NGS (BRCA-proficient); HRD / genomic instability (tumor) genomic instability PRESENT but qualitative only — multiple chromosomal imbalances / frequent LOH; BRCA-proficient; NO quantitative GIS/HRD score reported; MSI / MMR Microsatellite Stable (MSS); TMB Low (1 mut/Mb); PIK3CA amplification (somatic) focal amplification; MAP2K4 deletion (somatic) focal deletion; ER positive (~80% of tumor nuclei, weak-moderate intensity); PR negative; HER2 negative (IHC 1+); FOLR1 (folate receptor alpha) negative (<75% of cells at moderate/strong membranous intensity); CA-125 normalized on therapy (peak 940 -> 11.6 U/mL); Signatera ctDNA (MRD) cleared to Not Detected (30.3 -> 3.32 -> 0 MTM/mL)
- **Efficacy/toxicity weight:** 0.7
- **Toxicity vetoes:** additional peripheral neuropathy, severe myelosuppression / febrile neutropenia
- **Modality constraints:** oral or low-infusion-burden preferred, caution with agents carrying thrombotic or bleeding risk (history of high VTE risk; prior IVC filter)

6 ranked therapeutic options across 4 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Hrd Genomic Instability interventions

1. Rucaparib (1L maintenance)

Likelihood of effect: Moderate-to-high for PFS: replicated all-comer phase 3 (ATHENA-MONO ITT HR 0.52, 20.2 vs 9.2 mo) qualifying a BRCA-wild-type tumor without a BRCA or GIS gate.

Toxicity burden: Moderate (grade 3+ anemia 28.7%, neutropenia 14.6%, transaminase rise 10.6%)

Counter-productive MoA:Low (PARP myelosuppression is a patient-AE burden, not a mechanism that blunts the maintenance goal in genomically-unstable HGSC.)

Overall: Replicated all-comer PARP maintenance that qualifies a BRCA-wild-type tumor with no GIS gate; near-interchangeable with niraparib but lacks its tumor-specific pharmacology, and carries the same myelosuppression veto.

Key references: PMID 35658487 · NCT03522246

2. Niraparib (1L maintenance)

Likelihood of effect: High for PFS as an evidence fit: all-comer PRIMA phase 3 (HR 0.62, 13.8 vs 8.2 mo) plus BRCA-wild-type PDX pharmacology (pmid:30647846); deliverability gated by marrow reserve.

Toxicity burden: Moderate (grade 3+ thrombocytopenia 28.7%, anemia 31%, neutropenia 12.8%)

Counter-productive MoA: **Moderate** (Cytopenia-driven dose interruption on thin baseline marrow reserve can erode maintenance dose intensity; the advocate and conservative dissented on this toxicity-as-deliverability axis.)

Overall: Cleanest all-comer guideline and evidence fit with the only tumor-specific BRCA-wild-type pharmacology, but its grade 3+ thrombocytopenia/anemia hit the patient's named veto; deliverable only behind a documented hematology dose plan.

Key references: PMID 31562799 · PMID 30647846 · NCT02655016

3. Olaparib + bevacizumab (PAOLA-1 maintenance, contingent on validated HRD-positive)

Likelihood of effect: High ONLY if GIS ≥ 42 confirms HRD-positive: PAOLA-1 subgroup HR 0.33 (37.2 vs 17.7 mo); contingent and foreclosed below 42, ITT effect is a smaller HR 0.59.

Toxicity burden: High (combination grade 3+ AE ~57%; hypertension 19%, anemia 17%, up to 20% olaparib discontinuation)

Counter-productive MoA: **High** (Anti-angiogenic VTE/bleeding load stacked on PARP myelosuppression against an IVC-filter history; two vetoes stood on toxicity and evidence-quality grounds.)

Overall: Highest-magnitude maintenance if HRD-positive is confirmed, but contingent on a GIS she lacks and double-vetoes on stacked VTE/myelosuppression toxicity and a subgroup-only effect estimate.

Key references: PMID 31851799 · NCT02477644

Evidence in detail

Rucaparib (1L maintenance)

****Monk et al. J Clin Oncol 2022****

- **Disease context:** newly diagnosed advanced high-grade epithelial ovarian cancer, response after 1L platinum doublet (maintenance) — All-comers (4:1 randomization), HRD-positive and HRD-negative strata. Benefit held in the ITT population and in the non-nested HRD-negative subgroup, so a BRCA-WT, HRD-unquantified tumor still qualifies.
- **n:** 538
- **Toxicities:** anemia (grade ≥ 3): 28.7%; neutropenia (grade ≥ 3): 14.6%; ALT/AST elevation (grade ≥ 3): 10.6%; treatment discontinuation (AE) (grade any): 11.8%
- **Efficacy:** HR_PFS 0.52 HR (95% CI 0.4-0.68); HR_PFS 0.47 HR (95% CI 0.31-0.72); PFS 20.2 months

Niraparib (1L maintenance)

****Gonzalez-Martin et al. *N Engl J Med* 2019****

- **Disease context:** newly diagnosed advanced high-grade serous/endometrioid ovarian cancer, response after 1L platinum (maintenance) — All-comers, both HRD-positive and HRD-negative strata; no BRCA mutation and no validated HRD score required for enrolment. The all-comer enrolment is what makes PRIMA the cleanest fit for this BRCA-proficient, genomically-unstable tumor.
- **n:** 733
- **Toxicities:** anemia (grade ≥ 3): 31.0%; thrombocytopenia (grade ≥ 3): 28.7%; neutropenia (grade ≥ 3): 12.8%
- **Efficacy:** HR_PFS 0.62 HR (95% CI 0.5-0.76); HR_PFS 0.43 HR (95% CI 0.31-0.59); PFS 13.8 months; HR_OS 0.7 HR (95% CI 0.44-1.11)

Olaparib + bevacizumab (PAOLA-1 maintenance, contingent on validated HRD-positive)

****Ray-Coquard et al. *N Engl J Med* 2019****

- **Disease context:** newly diagnosed advanced high-grade serous ovarian cancer receiving 1L bevacizumab (maintenance) — All-comers enrolled, but the predefined and label-relevant benefit is in the HRD-positive subgroup (Myriad MyChoice GIS ≥ 42 or tumor BRCA mutation). This patient is BRCA wild-type and has only a qualitative Altera instability call, so eligibility is CONTINGENT on a validated GIS she does not yet have.
- **n:** 806
- **Toxicities:** hypertension (grade ≥ 3): 19%; anemia (grade ≥ 3): 17%; lymphopenia (grade ≥ 3): 7%; fatigue/asthenia (grade ≥ 3): 5%
- **Efficacy:** HR_PFS 0.59 HR (95% CI 0.49-0.72); HR_PFS 0.33 HR (95% CI 0.25-0.45); HR_PFS 0.43 HR (95% CI 0.28-0.66)

Vegf Angiogenesis interventions

1. Bevacizumab (continued maintenance)

Likelihood of effect: Moderate for PFS only: GOG-218 HR 0.717 (14.1 vs 10.3 mo), RoB2-low but no overall-population OS gain; smallest maintenance effect on the board.

Toxicity burden: Low (grade 2+ hypertension 22.9%, GI-wall disruption 2.6%; VTE/bleeding class caution)

Counter-productive MoA: Low (Anti-angiogenic wound-healing and VTE/bleeding risk is a patient-AE caution here, not a mechanism that blunts maintenance benefit; risktaker dissent was magnitude/preference-flavored.)

Overall: The only maintenance that clears both named vetoes on a backbone she already tolerates, but PFS-only with no OS gain and a live VTE/bleeding caution against her IVC-filter history.

Key references: PMID 22204724 · PMID 22204725 · NCT00262847

Evidence in detail

Bevacizumab (continued maintenance)

****Burger et al. *N Engl J Med* 2011****

- **Disease context:** newly diagnosed stage III-IV epithelial ovarian / primary peritoneal cancer (1L) — All-comers, no biomarker selection. Matches the VEGF/angiogenesis axis the patient is already being treated on.
- **n:** 1873
- **Toxicities:** hypertension (grade ≥ 2): 22.9%; **GI perforation/wall disruption** (grade any): 2.6%
- **Efficacy:** HR_PFS 0.717 HR (95% CI 0.625-0.824); **PFS** 14.1 months

****Perren et al. *N Engl J Med* 2011****

- **Disease context:** newly diagnosed epithelial ovarian / primary peritoneal cancer (1L) — All-comers, no biomarker gate. OS benefit concentrated in the high-risk subset (stage IV or suboptimally debulked); this stage IIIc R0 patient sits adjacent to that subset.
- **n:** 1528
- **Toxicities:** hypertension (grade ≥ 2): 18%
- **Efficacy:** HR_PFS 0.81 HR (95% CI 0.7-0.94); **OS** 36.6 months

II12 Tme interventions

1. GEN-1 / IMNN-001 (IL-12 immunogene, OVATION 2)

Likelihood of effect: Moderate but immature: OVATION 2 ITT PFS HR 0.79 (14.9 vs 11.9 mo) and OS HR 0.69 from sponsor topline; CIs unreported, regimen-matched to this patient.

Toxicity burden: Low (IP delivery reported well tolerated; no per-term grade 3+ table published)

Counter-productive MoA: Low (IL-12 immunogene acts via the peritoneal TME, sidestepping the foreclosed MSS/TMB-low checkpoint axis; no plausible counter-productive vector flagged.)

Overall: On-study IL-12 continuation that sidesteps the foreclosed checkpoint axis and carries a real OS signal, but efficacy is sponsor-topline-immature and delivery is intraperitoneal, not oral.

Key references: NCT03393884 · PMID 20033066

Er Expression interventions

1. Letrozole (endocrine maintenance)

Likelihood of effect: Low and weakly evidenced: only a retrospective ROBINS-I-serious series (60% vs 38.5% RFS at 24 mo, PMID:29157627); PR-negativity tempers the ER-driven signal.

Toxicity burden: Low (arthralgia, bone-density loss on prolonged aromatase inhibition; no named veto triggered)

Counter-productive MoA: Low (Endocrine blockade has no plausible counter-productive vector in ER-positive HGSC; the dissent here is evidence-quality, not mechanism.)

Overall: The lowest-toxicity oral option that clears every veto and fits the age band, but the evidence is a

single retrospective ROBINS-I-serious series and PR-negativity tempers the expected magnitude.

Key references: PMID 29157627

Evidence in detail

Letrozole (endocrine maintenance)

****Heinzelmann-Schwarz et al. *Gynecol Oncol* 2018****

- **Disease context:**ER-positive high-grade serous ovarian cancer, maintenance / later-line (maintenance) — ER-positive HGSOC by IHC; a translational study with a clinical letrozole-maintenance series, enriched for platinum-resistant and residual-disease patients. The patient is ER-positive (~80% nuclei) but PR-negative, which tempers expected magnitude.
- **n:** —
- **Toxicities:** No per-term toxicity table in the source; aromatase-inhibitor class toxicity is arthralgia and bone-density loss on prolonged use. Oral, low-burden, no named patient veto applies.
- **Efficacy:**RFS 60%; RFS 87.5%